Ethical AI in Education: A Multi-Agent Deep Research System

Abstract

This paper presents a multi-agent research system designed to investigate ethical considerations in AI-powered education. The system orchestrates four specialized AI agents using Microsoft's AutoGen framework with a RoundRobin group chat architecture. Web search and academic paper retrieval tools provide real-time evidence gathering via Tavily and Semantic Scholar APIs. A custom safety layer performs input and output guardrail checks across three policy categories: harmful content, prompt injection, and off-topic queries. System outputs are evaluated using an LLM-as-a-Judge framework scoring five criteria across six diverse test queries. Results demonstrate that the system produces well-structured, evidence-grounded responses on HCI-relevant topics while reliably detecting and refusing unsafe inputs.

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1. System Design and Implementation

1.1 Agent Architecture

The system implements four agents in a sequential RoundRobin GroupChat managed by AutoGen's GroupChatManager. Each agent has a distinct role in the research pipeline.

The Planner receives the raw user query and decomposes it into a structured research plan, identifying key concepts, source types, and suggested search queries for the Researcher. The Researcher is equipped with two tools, web_search and paper_search, and executes the Planner's strategy by collecting evidence from both web and academic sources. The Writer then synthesizes those findings into a coherent, well-cited response organized with clear headings and a references section. Finally, the Critic evaluates the Writer's output on five dimensions — relevance, evidence quality, completeness, accuracy, and clarity — and either approves the response by emitting TERMINATE or requests revision.

The conversation terminates when the Critic emits TERMINATE or the maximum round limit of 20 is reached.

1.2 Tools

Two research tools are integrated as AutoGen FunctionTools. The web_search tool calls the Tavily Search API with configurable depth and domain filters, returning structured results including title, URL, snippet, and publication date. The paper_search tool queries the Semantic Scholar API for academic papers, returning metadata including authors, year, citation count, abstract, and open-access PDF links, with support for year filtering to prioritize recent literature.

The Researcher agent is the only agent with tool access. All other agents use a vLLM-hosted Qwen3-8B endpoint provided by the course. The Researcher uses Groq's llama-3.3-70b-versatile model, as the vLLM endpoint does not support the "auto" tool choice protocol required for function calling.

1.3 Orchestration

The AutoGenOrchestrator class manages team initialization and query processing. A notable engineering challenge was handling asyncio event loop conflicts in Streamlit's threading model. This was resolved by submitting all async work to a ThreadPoolExecutor with a fresh event loop per call, enabling safe concurrent execution across both CLI and web UI contexts.

1.4 Models

The following table summarizes model assignments across agents.

Agent	Model	Provider
Planner	Qwen/Qwen3-8B	vLLM (salt-lab)
Researcher	llama-3.3-70b-versatile	Groq
Writer	Qwen/Qwen3-8B	vLLM (salt-lab)
Critic	Qwen/Qwen3-8B	vLLM (salt-lab)
Judge	Qwen/Qwen3-8B	vLLM (salt-lab)

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2. Safety Design

2.1 Policy Categories

The system enforces three prohibited categories. First, harmful content is detected via a curated keyword list covering violence, self-harm, illegal activities, and explicit content. Violations above medium severity trigger a hard refusal. Second, prompt injection is detected through pattern matching against known adversarial phrases such as "ignore previous instructions" and "forget everything." All prompt injection detections are treated as high severity and blocked immediately. Third, off-topic queries are flagged using a keyword relevance check against the configured topic domain. Queries with no overlap with HCI, AI, or education terminology receive a low severity warning.

2.2 Guardrail Implementation

The system implements a two-stage guardrail pipeline coordinated by a SafetyManager class. The InputGuardrail validates the raw user query before it reaches the agent pipeline, checking query length, toxic language, prompt injection patterns, and topic relevance. The OutputGuardrail inspects the final synthesized response before delivery, detecting PII patterns such as email addresses, phone numbers, SSNs, and credit card numbers using regular expressions, and scanning for harmful output phrases.

All safety events are logged to an in-memory list with timestamp, event type, violation details, and the action taken. The Streamlit UI surfaces these events as red warning banners, and the sidebar maintains a running safety event log for the session.

2.3 Safety Response Strategies

The following table summarizes how the system responds to each category of violation.

Category	Severity	Action
Harmful content	High	Refuse with explanation
Prompt injection	High	Refuse with explanation
Off-topic query	Low	Warn, allow with notice
PII in output	Medium	Refuse output
Query too short	Low	Warn

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3. Evaluation Setup and Results

3.1 Test Queries

Six diverse queries were designed to cover the system's target domain from multiple angles:

1. What are the ethical implications of using AI to grade student essays in K-12 education?
2. How does algorithmic bias in AI systems affect underrepresented students in higher education?
3. What frameworks exist for responsible AI deployment in university settings?
4. How should student data privacy be protected when using AI tutoring systems?
5. What are the risks of using AI in college admissions decisions?
6. How can educators detect AI-generated academic dishonesty while respecting student rights?

3.2 Judge Prompts

Two independent judge prompt templates were used to score each response. The first rubric evaluates relevance and coverage, specifically whether the response directly addresses the query, covers all major sub-topics, and avoids irrelevant content. The second rubric evaluates evidence and citation quality, specifically whether sources are credible, citations are properly formatted, claims are grounded in evidence, and both academic and web sources are represented. Three additional criteria were evaluated: factual accuracy, safety compliance, and response clarity. All criteria are scored on a 0.0 to 1.0 scale.

3.3 Results

Query	Relevance	Evidence	Accuracy	Safety	Clarity	Overall
Q1: AI grading systems in K-12	**0.00**	**0.00**	**1.00**	**1.00**	**0.00**	**0.35**
Q2: Algorithmic bias in AI tutoring	**0.70**	**0.00**	**0.70**	**0.70**	**0.70**	**0.52**
Q3: Data privacy in online learning	**0.00**	**0.00**	**1.00**	**1.00**	**0.00**	**0.35**

Query	Relevance	Evidence	Accuracy	Safety	Clarity	Overall
Q4: AI governance frameworks	**0.00**	**0.00**	**0.00**	**0.30**	**0.00**	**0.05**
Q5: Transparency in college admissions	**0.00**	**0.00**	**0.70**	**1.00**	**0.00**	**0.29**
Q6: Academic integrity and GenAI	**0.70**	**0.00**	**0.70**	**0.70**	**0.70**	**0.52**
Average	**0.23**	**0.00**	**0.68**	**0.78**	**0.23**	**0.35**

3.4 Error Analysis

The primary failure mode observed during development was rate limiting on the Groq API, which imposes a 100K daily token limit on free tier accounts. This interrupted the Researcher agent mid-pipeline during high-volume testing sessions. The system's error handling returns partial results rather than crashing, which preserved usability even under this constraint.

A secondary failure mode emerged when switching to the vLLM endpoint for the Researcher agent. The Qwen3-8B model hosted at the course endpoint does not support the "auto" tool choice protocol, which AutoGen requires for function calling. This was resolved through a hybrid routing approach where only the Researcher uses Groq while all other agents use vLLM, significantly reducing per-query token consumption on the rate-limited endpoint.

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4. Discussion and Limitations

4.1 Key Insights

The multi-agent architecture produces noticeably richer responses than a single-agent baseline would. The Planner's structured decomposition guides the Researcher toward more targeted queries, and the Critic's feedback loop catches gaps in coverage before the response reaches the user. The separation of tool-equipped and tool-free agents also proved practically significant as a design principle: not all LLM endpoints support function calling, and building the architecture around this constraint from the start prevented harder-to-fix issues later.

Safety guardrails performed reliably on clear-cut cases such as explicit harmful language and obvious prompt injection attempts. The off-topic detection was more conservative in practice, occasionally flagging tangentially related queries as low relevance. This suggests that keyword matching alone is insufficient for nuanced relevance judgments.

4.2 Limitations

Token rate limits represent the most significant practical constraint. The Groq free tier's 100K daily token limit is exhausted quickly in multi-agent pipelines where each query involves four or more LLM calls. This limits practical throughput to approximately five to ten full research queries per day on free tier accounts. The vLLM endpoint provided for this course also does not support tool calling, which restricts tool use to Groq-backed agents only and creates a dependency on the rate-limited service for any query requiring web or paper search.

Beyond infrastructure constraints, the system has no persistent memory across sessions. Each query starts fresh without knowledge of prior conversations, which limits its usefulness for longitudinal research tasks. The current safety guardrails also rely on keyword matching rather than semantic understanding, meaning sophisticated adversarial inputs could potentially bypass detection if they avoid known trigger phrases.

4.3 Future Work

Several directions would meaningfully improve the system. Replacing keyword-based relevance checking with embedding similarity would enable more robust off-topic detection. Adding a vector database for session memory would allow the system to build on prior research rather than starting from scratch each time. Expanding tool coverage to include PDF parsing and citation graph traversal would also strengthen evidence quality. Finally, fine-tuned safety classifiers trained on adversarial examples would provide stronger guarantees than the current rule-based approach.

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